

Design Thinking, Innovation and Entrepreneurship (25HU102)

Question Bank

Unit-3 Design Thinking Techniques

Short Questions & Answers

1. **Q:** What is the main focus of Design Thinking tools?
A: To support creativity and problem-solving centered around user needs.
2. **Q:** What is meant by “user-centered” in Design Thinking tools?
A: It means the tools focus on understanding and solving real problems faced by users.
3. **Q:** Why are Design Thinking tools called iterative?
A: Because they allow continuous improvement through repeated testing and feedback.
4. **Q:** Which tools are used in the *Empathize* stage?
A: Empathy Map, Interviews, Observation, and Persona Creation.
5. **Q:** Which tools are used in the *Define* stage?
A: Point of View (POV) Statement, Problem Statement, and Affinity Diagram.
6. **Q:** What is the purpose of *Ideation tools*?
A: To generate a wide range of creative ideas and possible solutions.
7. **Q:** Which tools are commonly used in the *Prototype* stage?
A: Sketching, Storyboarding, and 3D Modeling.
8. **Q:** What is the main function of *Testing tools* in Design Thinking?
A: To gather user feedback and evaluate how well a solution works.
9. **Q:** Name one digital tool used for prototyping.
A: Figma or Adobe XD.
10. **Q:** What is empathy in Design Thinking?
A: Empathy is the ability to understand and share the feelings, thoughts, and experiences of users.
11. **Q:** What is the purpose of listening in the empathy stage?
A: To gather deep insights into users’ experiences, emotions, and challenges.
12. **Q:** What is *empathy mapping*?
A: A tool used to visualize what users *say, think, feel, and do* to better understand their perspective.
13. **Q:** What is *shadowing* in empathy research?
A: Observing users in their real environment to see how they interact with products or services.
14. **Q:** What is the main goal of the Define stage?
A: To frame a clear and actionable problem statement that guides the ideation process.
15. **Q:** What key tool is used in the Define stage?
A: The *Point of View (POV) statement*.
16. **Q:** How does the Define stage connect to Ideation?
A: A well-defined problem provides direction and inspiration for creative idea generation.
17. **Q:** What is the main goal of ideation?
A: To explore a wide range of possible solutions before selecting the best one.
18. **Q:** Name any three ideation techniques.
A: Brainstorming, Mind Mapping, and SCAMPER.
19. **Q:** What is *brainstorming*?
A: A group activity where participants share ideas freely without criticism to encourage creativity.
20. **Q:** What is *mind mapping*?
A: A visual technique that organizes ideas around a central theme using branches and keywords.

21. **Q:** What does *SCAMPER* stand for?
A: Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse.
22. **Q:** What is *storyboarding* used for in ideation?
A: To visualize the user journey or process through a sequence of sketches or images.
23. **Q:** What are the two main forms of prototyping?
A: Low-fidelity prototyping and high-fidelity prototyping.
24. **Q:** What is *low-fidelity prototyping*?
A: Simple, rough, and inexpensive models such as sketches, paper mock-ups, or storyboards used to test early ideas. Eg: Paper sketches, wireframes, or cardboard models.
25. **Q:** What is *high-fidelity prototyping*?
A: More detailed and interactive models that closely resemble the final product in look and function. Eg: A digital app mock-up or a 3D-printed product sample.
26. **Q:** Why is prototyping important in Design Thinking?
A: It helps test ideas quickly, identify problems early, and improve the final solution based on user feedback.
27. **Q:** At which stage of Design Thinking does testing occur?
A: It occurs at the final stage but is also iterative throughout the process.
28. **Q:** What is *usability testing*?
A: Observing users as they interact with a product to check ease of use and satisfaction.
29. **Q:** What is *A/B testing*?
A: Comparing two design versions to see which performs better with users.
30. **Q:** What is *concept testing*?
A: Presenting an idea to users to get early feedback before building a prototype.
31. **Q:** What is the main goal of testing in Design Thinking?
A: To validate assumptions and ensure the design meets user needs.
32. **Q:** Why is user feedback important during testing?
A: It helps designers understand real-world problems and refine solutions effectively.
33. **Q:** What is the main objective behind launching Pureit?
A: To provide safe and affordable drinking water to households, especially in rural and low-income areas.
34. **Q:** Which company introduced the Pureit water purifier?
A: Hindustan Unilever Limited (HUL).
35. **Q:** What is Pureit's unique selling proposition (USP)?
A: 100% safe water without the need for electricity or continuous water supply, at an affordable cost.
36. **Q:** Who is the target market for Pureit?
A: Middle- and lower-income households in both rural and urban India.
37. **Q:** Who is the parent company of Royal Enfield and who played a key role in reviving the brand?
A: Eicher Motors Limited, Siddhartha Lal (CEO of Eicher Motors).
38. **Q:** What segment of customers does Royal Enfield primarily target?
A: Middle- and upper-middle-class youth who value adventure and retro style.
39. **Q:** What major strategy helped Royal Enfield regain its market position?
A: Repositioning the brand as a lifestyle and adventure motorcycle brand.
40. **Q:** Name one popular modern model of Royal Enfield.
A: Classic 350 / Himalayan / Interceptor 650.

41. **Q:** What type of business model does Big Basket follow?
A: Online grocery delivery (e-commerce).
42. **Q:** What was the main aim of launching Big Basket?
A: To provide customers with a convenient and reliable way to buy groceries online.
43. **Q:** What delivery options does Big Basket provide?
A: Same-day delivery, next-day delivery, and express 90-minute delivery (BB Express).
44. **Q:** What problem was IDEO trying to solve with the water pump project?
A: To design an affordable and user-friendly water pump for farmers in developing countries to improve irrigation and reduce manual labor.
45. **Q:** What is the main feature of the IDEO-designed water pump?
A: It is a low-cost, manually operated treadle pump that can draw water efficiently without electricity.
46. **Q:** What is the core idea behind Airbnb's business model?
A: Enabling people to rent out their homes or rooms to travelers through an online platform
47. **Q:** What type of business model does Airbnb use?
A: A peer-to-peer (P2P) sharing economy model.
48. **Q:** How does Airbnb generate revenue?
A: By charging service fees from both hosts and guests for each booking.
49. **Q:** How did the COVID-19 pandemic affect Airbnb's business?
A: It led to a drop in international travel but boosted demand for local and long-term stays.
50. **Q:** What type of products does Oral-B manufacture?
A: Oral care products such as toothbrushes, toothpaste, and electric toothbrushes.
51. **Q:** Which company owns Oral-B today?
A: Procter & Gamble (P&G).
52. **Q:** What kind of innovation is Oral-B known for in recent years?
A: Smart electric toothbrushes with AI-based brushing feedback.

Long / Analytical Questions and Answers

- 1. In what ways do the core characteristics of Design Thinking tools differentiate them from traditional problem-solving methods?**
Hint: Compare traditional linear approaches with the iterative, user-focused nature of Design Thinking.
- 2. Explain the different types of Design Thinking tools used at each stage of the process with suitable examples.**
Hint: Describe tools for Empathize (Empathy Map, Persona), Define (POV Statement, Affinity Diagram), Ideate (Brainstorming, Mind Mapping), Prototype (Sketching, Storyboarding), and Test (Usability Testing, Feedback).
- 3. Compare and contrast the tools used in the Ideation and Prototyping stages of Design Thinking.**
Hint: Highlight differences in purpose—creative idea generation versus tangible model creation—and give examples.
- 4. Explain the characteristics of effective Design Thinking tools and their contribution to innovation.**
Hint: Discuss features such as user-centeredness, flexibility, collaboration, and visual communication.
- 5. Explain the importance of listening and empathizing techniques in the Design Thinking process.**

Hint: Discuss how empathy helps designers understand user needs, build trust, and design human-centered solutions.

6. **Describe the various techniques used to develop empathy with users in Design Thinking.**
Hint: Include interviews, observation, shadowing, empathy mapping, and storytelling.
7. **Evaluate the role of user observation and shadowing in empathizing with users.**
Hint: Explain how real-world observation uncovers hidden pain points and unmet needs.
8. **Explain how storytelling can be used as a tool for empathy in Design Thinking.**
Hint: Discuss how sharing user stories helps teams connect emotionally and design meaningful solutions.
9. **Explain the importance of the Define stage in the Design Thinking process. How does it influence the success of later stages?**
Hint: Discuss how defining the right problem based on user insights ensures focused ideation and effective solutions.
10. **Describe the key activities and outcomes of the Define stage in Design Thinking.**
Hint: Include data synthesis, creation of user personas, problem framing, and development of a Point of View (POV) statement.
11. **Analyze the role of the Point of View (POV) statement in the Define stage and its impact on the design process.**
Hint: Explain how the POV helps narrow focus and aligns the design team around user needs.
12. **Explain the importance of ideation in the Design Thinking process and describe various ideation techniques with examples.**
Hint: Discuss how ideation bridges problem definition and prototyping; include techniques such as brainstorming, mind mapping, SCAMPER, and storyboarding.
13. **Describe the SCAMPER technique and analyze how it helps in generating innovative solutions.**
Hint: Explain each step—Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse—with examples.
14. **Discuss how ideation techniques support collaboration and divergent thinking in multidisciplinary teams.**
Hint: Explain the importance of open-mindedness, empathy, and cross-functional inputs.
15. **Critically analyze the connection between ideation and prototyping in Design Thinking.**
Hint: Explain how ideation outcomes feed directly into tangible prototypes for testing and feedback.
16. **Explain the different forms of prototyping in Design Thinking and discuss their significance in the innovation process.**
Hint: Describe low-fidelity and high-fidelity prototypes, their characteristics, uses, and how they help refine ideas through user feedback.
17. **Discuss how low-fidelity and high-fidelity prototyping complement each other in the Design Thinking process.**
Hint: Compare their cost, time, and feedback value, showing how teams move from simple sketches to realistic prototypes.
18. **Critically evaluate how prototyping reduces risks and improves decision-making in product or service design.**
Hint: Explain how prototypes help identify usability issues, validate concepts, and prevent costly design errors.
19. **Explain the different forms of testing used in the Design Thinking process and their importance in product or service development.**
Hint: Discuss usability testing, prototype testing, A/B testing, pilot testing, and concept testing. Explain how each helps in validating user needs and improving design solutions.
20. **Discuss the various methods of testing in Design Thinking with suitable examples.**
Hint: Include methods such as user observation, interviews, role-playing, low-fidelity and high-fidelity prototype testing, and digital simulations.

21. **Analyze how early-stage and late-stage testing differ in Design Thinking. Why are both essential for a successful design?**
Hint: Contrast low-fidelity (paper/mock-up) testing with high-fidelity (final prototype/pilot) testing in terms of cost, speed, and insights.
22. **Discuss the key factors that led Hindustan Unilever Limited (HUL) to launch Pureit.**
Hint: Explain the social and economic issues related to access to safe drinking water, HUL's CSR goals, and market opportunities in India.
23. **Critically analyze the overall impact of Pureit on public health and rural development in India.**
Hint: Discuss outcomes such as reduced waterborne diseases, consumer empowerment, and social awareness.
24. **Explain how Royal Enfield transformed from a struggling motorcycle manufacturer into a global lifestyle brand.**
Hint: Discuss product innovation, marketing, customer engagement, and expansion strategies.
25. **Discuss how Royal Enfield maintained a balance between heritage and modernity in its product design and brand image.**
Hint: Explain how the company kept its classic appeal while introducing new technology and global standards.
26. **Explain Royal Enfield's global expansion strategy and its impact on brand image.**
Hint: Include international store launches, exports, and global events targeting motorcycle enthusiasts.
27. **Analyze the key factors that contributed to the success of Big Basket in the Indian online grocery market.**
Hint: Include points such as convenience, technology, supply chain management, customer trust, and market timing.
28. **Examine the role of technology and innovation in Big Basket's operations and customer engagement.**
Hint: Mention AI-driven demand forecasting, app-based ordering, route optimization, and personalized offers.
29. **Discuss how IDEO applied the principles of human-centered design (HCD) in developing the water pump.**
Hint: Explain how IDEO involved users in every stage — research, ideation, prototyping, and testing — to create a solution suited to local farmers' needs.
30. **Discuss how IDEO's approach to the water pump reflects the broader concept of "design for social innovation."**
Hint: Show how design thinking can be applied to address global challenges like poverty, water scarcity, and rural development.
31. **Discuss how Airbnb disrupted the traditional hospitality industry.**
Hint: Explain how Airbnb's peer-to-peer model, technology platform, and community-driven approach challenged hotels on price, experience, and flexibility.
32. **Discuss the challenges Airbnb faces in regulatory compliance and community acceptance.**
Hint: Explain zoning laws, housing affordability concerns, neighbourhood issues, and city-level restrictions.
33. **Analyze Airbnb's strategies for building customer trust and loyalty in a peer-to-peer environment.**
Hint: Mention verification systems, reviews, insurance, customer service, and the "Belong Anywhere" philosophy.
34. **Explain how Oral-B differentiates itself from competitors like Colgate and Philips Sonicare.**
Hint: Compare technology, dentist endorsements, product design, and brand messaging.
35. **Discuss how Oral-B's association with Procter & Gamble (P&G) has influenced its global expansion and innovation.**
Hint: Include the impact of P&G's distribution network, research capabilities, and marketing strength.